

RetroHabits

A personal assistant for students: your schedule, habits and memory in one iPhone app, showing how a change in one part of your week affects the rest.

What it does

Feature	What you see
Agenda home	Date circles for the next two weeks on top, this week's budget line, then one day's plan: Canvas, iPhone Calendar (incl. Teams meetings) and email deadlines, each rewritten by Gemini as one line. Below: a Classes card (that day's classes plus tomorrow's) and every habit still to do as a checkbox.
Habits	Tick a habit and it slides off the list, today's reminder for it is cancelled, and the tick is recorded for Ask. Undo in the Routine tab, which also holds classes, shifts and bills.
Class schedule	From the + button: add, edit or swipe-to-delete classes. Each runs for a four-month term by default (an editable end date) and shows on its day and the day before, with a reminder before it starts.
Ripple Mode	Say what changed ("my lab moved to Friday evening") or ask "what if…". See what it collides with, free slots for displaced plans, what it would cost against this week's budget, and accept or reject a revised plan.
Campus Capture	Photograph an event flyer. Get the event details, calendar fit, which of your shared interests it matches, what the flyer leaves out, and whether its price fits that week's budget.
Money	A weekly budget (Monday to Sunday). Log spending by typing, dictation or a receipt photo; with no amount stated it asks "How much was it?" instead of guessing. Shows planned shift pay and upcoming bills (both planned, not counted), spending by category and an 8-week trend.
Voice log + recall	Record "leg day, squats 3×8 at 135". It's logged as structured data; on the same weekday next time the app shows what you did and suggests repeat / progress / ease off.
Ask	Chat by typing or tapping the mic (on-device speech-to-text). Answers use your agenda, logs, email, budget, long-term memory and 120 days of history, so "when did I submit my normalization lab?" gets a date (or "I don't have any record"). Replies can be read aloud in the ElevenLabs voice.
Memory Inspector	Tap *Why?* on any suggestion to see the memories it used, then correct or forget them; the suggestion updates.
Inbox & email deadlines	Recent Gmail (incl. Teams notifications) as tappable one-liners with Open in Gmail. Deadlines and meetings an email states become agenda items with alerts, and Ripple checks them for conflicts.
Spoken briefing	One tap reads today's plan, urgent email and training aloud (Gemini writes it, ElevenLabs speaks it).

How it fits together

iPhone app (SwiftUI) → **backend** (FastAPI on a Vultr server, HTTPS via Caddy, bearer-token auth) → **Gemini** (summaries, voice, flyer and receipt understanding, chat) · **Tiger Data** (Postgres; hypertables for logs and spending, summary cache, history) · **Backboard** (long-term memory) · **ElevenLabs** (spoken briefing) · **Gmail API** (read-only).

- **No invented connections.** In Ripple and Capture, Gemini only parses input and must list its assumptions; conflicts and free slots come from plain overlap code (`backend/ripple.py`).
- **You stay in control.** Nothing changes until you press Accept; the app never edits your calendar, and a moved session is never marked done.
- **Money is never guessed.** Recorded, planned (Ripple, Capture, shift pay, bills) and hypothetical amounts are kept apart; a missing amount is asked for.
- **Resilient AI.** Gemini calls fall back across six models (the free tier allows ~20 requests/day per model) with a 25 s timeout per model.

- **Calm, native feel.** Warm charcoal and crimson in dark mode (light mode too), swipe between tabs, a sliding dock with a + menu for Class Schedule, Stats and Setup, and Dynamic Type throughout.

Run it

Backend (already deployed at `https://45-76-234-93.sslip.io`). To run locally:

```
cd backend && python3 -m venv .venv && .venv/bin/pip install -r requirements.txt
cp .env.example .env # fill in the keys
.venv/bin/python init_db.py && .venv/bin/uvicorn main:app --port 8000
```

iPhone: plug in the phone, then run the script below. It builds, installs and launches the app with the server token baked in (the token stays out of git). Free Apple accounts expire after 7 days — just run it again.

```
scripts/phone.sh
```

Deploy backend changes:

```
backend/deploy.sh # code only
backend/deploy.sh --env # code + .env (+ Gmail token)
```

Configuration (`backend/.env`)

Key	Where to get it
DATABASE_URL	Tiger Cloud service connection string
GEMINI_API_KEY	aistudio.google.com → API keys
BACKBOARD_API_KEY	app.backboard.io → API Keys
ELEVENLABS_API_KEY	elevenlabs.io → Developers → API keys (starts with sk_)
API_TOKEN	Any long random string; the app sends it to the server
Gmail (optional)	Desktop OAuth client → backend/google_client.json; add yourself as a test user; enable Gmail API; run gmail_auth.py once (read-only), then deploy.sh --env

Demo in 3 minutes

- `backend/.venv/bin/python backend/demo_seed.py` — two weeks of history and sample memories (`--clear` removes them).
- In the app: create a **Gym** habit on Fridays, reminder 6:00 PM, 90 min.
- **Capture** a robotics-workshop flyer → it matches your interest, flags the Friday gym clash and the missing price.
- **Plan around it** → Ripple shows the gym conflict and free slots → **Accept**.
- Tap **Why?** → correct "I like robotics" to "that was my roommate" → re-check updates the verdict.

Known limits

- Gemini free quota is ~120 requests/day across models; enable billing before a heavy demo. No bank connection: spending is logged by you.
- A Canvas .ics feed doesn't say what you turned in: tap *Mark as Turned In* (a Canvas API token records submissions automatically). History is recorded from now on, and a record made offline isn't retried.
- Gmail sign-in in Testing mode expires after 7 days (rerun gmail_auth.py and deploy.sh --env, or publish the consent screen).
- Builds from Xcode's Run button don't include the server token; use scripts/phone.sh or paste API_TOKEN in Setup. The widget target must still be added (WIDGET-SETUP.md).